BTS-II(R&S)/I(S)-04.18-0818 Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ]

B

B.Tech. Degree II Semester Regular and Supplementary/ I Semester Supplementary Examination April 2018

CS/EC/IT AS 15-1202 B & CE/EE/ME/SE AS 15-1102 A
ENGINEERING PHYSICS
(2015 Scheme)

Time: 3 Hours Maximum Marks: 60

PART A
(Answer *ALL* questions)

$(10 \times 2 = 20)$

I. (a) Sketch the three essential requirements of a laser system, explain them briefly.
(b) Discuss the recording of Holographic images using a He-Ne laser.
(c) Explain why there is not graded index single mode optical fiber.
(d) What do you mean by packing factor? Calculate packing factor for simple cubic, BCC and FCC structures.
(e) Discuss any two properties of metallic glasses.
(f) What do you mean by pseudo elasticity in shape memory alloys?
(g) Classify nanomaterials with appropriate examples.
(h) What is Meissner effect in superconductors?
(i) What do you mean by a black body? Mention "Plank's formula" for black body radiation.
(j) State and explain Heisenberg's uncertainty principle.

PART B

$(4 \times 10 = 40)$

II. (a) With the help of the energy band diagram discuss working of a diode laser. (8)
(b) A three level laser system emits light of wavelength 600 nm. At what temperature the ratio of population of the upper level to the lower is equal to half? (2)

OR

III. (a) What is the principle underlying the working of optical fibres? Derive an expression for numerical aperture of a step index fibre. (8)
(b) The velocity of light in the core of a silica fiber 2×10^8 m/s and the critical angle of core-cladding interface is $60°$. Calculate: (2)
(i) The refractive indices of core and cladding.
(ii) The numerical aperture of the fiber.

(P.T.O)

IV. (a) List the seven crystal systems with lattice parameters and number of unit cells in each case. (7)

(b) A beam of electron accelerated with 120 V is used for diffraction experiment in an fcc crystal. If the first order diffraction from the (111) plane occurs at 22°, what is the lattice parameter of the crystal. (3)

OR

V. (a) Describe the construction and working of liquid crystal display system with a neat diagram. What are the merits and demerits of the LCD? (8)

(b) What are shape memory alloys? What are the applications of shape memory alloys? (2)

VI. (a) What are superconductors? Discuss the following terms. (8)

- (i) dc-Josephson effect.
- (ii) ac-Josephson effect.
- (iii) Quantum interference.
- (iv) SQUID.

(b) What do you mean by top down and bottom up approach of nanomaterial synthesis? (2)

OR

VII. (a) Derive Einstein's mass energy relationship. (8)

(b) How fast must an electron move in order to have its mass equal to the rest mass of proton (1.67×10^{-27} kg)? (2)

VIII. (a) Derive time dependent Schrodinger equation. What is the significance of wave function in quantum mechanics? (8)

(b) Calculate the energy difference between the ground state and first excited state for an electron in a box of length of 5 nm. (2)

OR

IX. (a) What is magnetostriction? Discuss the construction and working of ultrasonic generator based on magnetostriction effect. (7)

(b) What is the reverberation time of a cubic hall if a false ceiling is made at half the height of the hall (at 7 feet)? The absorption coefficient of the false ceiling material is 0.3 and that of wall and floor is 0.25. Analyzing your result, suggest suitable actions needed (if any) to maintain acoustic efficiency. (1 foot = 0.3048 m). (3)

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